Final Project: Pick a Project

Computer Science - Week 10

Oct 3, 2026 - Version 0.0.2

Please make sure to your UD **email** AND **name** is below. Unless your UD emails are included in this table, then you will not earn any points for this assignment when it is graded! **Every person should submit their own individual worksheet**.

Talk about each of the questions with a friend or our LLM (or both), but **make sure the words you enter below are yours** (with the obvious exceptions notwithstanding). Brainstorming is best done in full sentences, and the main goal here is brainstorming. It's worth reviewing the information in this series of pages in the Drafter documentation to gain a better understanding of the process.

Role	Name	Email
Developer (You)	Ashleigh Cheeseman	ascheese@udel.edu
Friend (optional)		

Part 1: The Goal & Prior Implementations

1. **Identify the Goal:** What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The latter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major.
    - **Biomedical Engineering** majors may consider one of the following:
        - If you have started (or completed) the Cell/Tissue Design final project in BMEG100, design an organoid, which involves combining a cell type, biomaterial, and testing method. Have the website include a description and diagram of your design concept, a testing method, a needs statement, and an impact statement relevant to their organoid. Also include a link to a paper that supports your idea and a paragraph describing how that paper supports your design concept.

        - Otherwise, create a website related to your BMEG100 Lab Report 1, *Cytoskeletal and Nuclear Staining Lab Report Guidelines*, linked here (University of Delaware account required to access).
    - **Chemistry** and **Chemical Engineering** majors may consider creating an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e (OpenStax) — Chapter 4: Chemical Reactions and Stoichiometry. To avoid manually compiling atomic mass tables or writing custom string tokenizers, use the ChemPy library to parse molecular formulas into exact molecular weights and verify reaction stoichiometry against compound data from the NIH PubChem Database. The program outputs an organized breakdown of limiting reagent status, product yields, and remaining masses of unreacted species. Also use the heat of the reaction to determine if it is exothermic or endothermic, and compute the heat flow per mole of the product.
    - Majors in the **Electric & Computer Engineering** department may consider a steganography project that hides messages in images, and then retrieves those messages from uploaded images.
    - **Mechanical Engineering** majors may consider a project that somehow uses both the NumPy and SciPy Python libraries.

Take in characteristics of birds and return with possible birds in the area

2. **Prior Implementations:** Search the internet. What existing tools or websites currently attempt to address it?

There are different apps and websites that allow for users to input characteristics and get a list of possible birds in return, like All About Birds or Merlin Bird ID.

3. **Your Implementation:** Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

I would want this web application to be used by everyday people, not experts, to be able to identify common birds that they could see in their backyard or around the neighborhood.

Part 2: The Input-Process-Output (IPO) Model

Think about the fundamental flow of your program.

1. **The Core Goal:** Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

This program will take user input on a few characteristic fields with specific options and then process that data to return a list of possible species they could've seen. In this, the program will return a list of best matches, not only birds that match exactly all characteristics, and the user will also be able to leave some characteristics blank in their initial search.

2. Input: What specific information will the user need to provide to your program? (e.g., a city name, a numerical budget, a list of hobbies).

Size, main color(s) (could give list), location/habitat (ex:forest, bush, plains, etc), diet, and/or other distinct features

3. Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).

The computer will use the inputs to search through a set of bird species with these characteristics filled out- likely a dataclass with designated fields for each characteristic. It will find birds that match or have some of the characteristics that the user input

4. Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).

The user will get back possible options that match perfectly or match somewhat with their inputs.

Part 3: Data Abstraction

Identify the "nouns" of your problem.

1. **Representing Reality:** Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things"

(nouns) your program needs to track? (e.g., *If building a recipe app, the nouns are "Recipe" and "Ingredient"*).

Bird(s), Colors, Habitat, Diet, Features

2. **Identifying Properties:** For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?
    - *Details I need:* (e.g., *Ingredient name, weight, cost*)
    - *Details I will ignore:* (e.g., *Ingredient color, brand, expiration date*)

I need- color and habitat mostly I will probably need diet and features I will ignore features that are not distinct enough or not applicable

Part 4: Challenges, AI Strategy, & Goals

1. **Difficulties:** What parts of this project seem the most difficult or technically complex?

I will have to make a data structure of the possible birds, and making it so the characteristics do not have to match completely.

2. **Coding Strategy:** If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project.

I will ensure I truly understand the generated code by making sure I read through it thoroughly and ask the LLM questions if I get lost. I will also make sure that I understand all the concepts and tactics used to generate the code.

3. **Goals:** What is your ultimate goal for this project? (e.g., *Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply*).

My ultimate goal is to make a program that I myself would enjoy using and to better understand drafter and how it is used in the world.

Part 5: Research & Brainstorming Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. **The Initial Prompt/Question:** Ask the AI or your friend to help you brainstorm. (e.g., *"I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"*). Paste your prompt below:

I want to build a web tool that will help everyday people identify birds and help people appreciate the nature they see everyday.

2. **The Response:** Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?

I know of a few apps that already exist for this problem as I have used them before, and those are very large scale search engines.

3. **The Pushback:** You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain why it is too complicated for an introductory programming project. Ask them for a simpler alternative. *Paste your response here:*

When brainstorming, the LLM noted the difficulty of identifying a bird based on an image in the context of this course, so I will be limiting my project to only text inputs and also narrowing the possible options to a very limited number of species (~20).

4. **Reflection:** Based on this conversation, how has your project idea changed or become more focused?

My project has become more focused because it will only be able to identify the most common species seen in the area and will return some additional information instead of just the name of the species.

Part 6: Resource Planning

1. What resources will you use if you get stuck?

If it get stuck, I will ask an LLM or a friend to help me work through my struggles.

2. If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

I will trace through the code to see where the problem is and if I cannot figure out how to fix it on my own, I will show the LLM the error message to see if it has any suggestions for me.

Part 7: LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

would it be reasonable to make a program that serves as a search engine for birds? it would have user inputs of different characteristics and return a list of possible birds

Yes! A bird search engine would be a very reasonable project for a CS1 final project using Drafter. It's a good idea because it has a clear purpose, takes user input, processes a collection of data, and returns useful results. It also gives you room to make the application interesting without making it too complicated.

I'd recommend building a bird identification and discovery tool that lets users select characteristics and see which birds match.

What the application could look like



1. User selects bird characteristics

The user fills in options such as:

- Bird size: small, medium, or large
- Main color: blue, red, brown, black, etc.
- Habitat: forest, wetlands, backyard, ocean
- Diet: seeds, insects, fish, or other



2. Program searches its bird collection

The program compares the selected characteristics with the properties stored for each bird. It filters out birds that don't match the user's selections.



3. Results show possible matches

Each result could display:

- Bird name
- Picture or emoji
- Size, color, and habitat
- A short description or fun fact

Is it realistic with Drafter?

Absolutely. The key is to build a search tool using a collection of bird data, rather than trying to build a full internet search engine.

Feature	Difficulty	Recommendation
Dropdowns or checkboxes for characteristics	Easy	Include
Filter birds based on selections	Easy–moderate	Core feature
Display matching birds and descriptions	Easy	Include
Show bird pictures	Moderate	Optional enhancement

Save favorite birds using local storage	Moderate	Optional enhancement
Identify a bird from an uploaded photo using AI	Difficult	Avoid for the initial version
Search the entire internet for bird information	Difficult	Avoid for the initial version

You can implement the core functionality in ordinary Python. Drafter can handle the user interface, and you can store bird information in a Python list of dictionaries. You do not need a backend or database.

How the program would work (IPO model)

This matches the Input-Process-Output structure in your project planning worksheet.

CS1 Final Project Alpha F26.pdf

Input

Color, size, habitat, diet

Processing

Compare user selections against stored bird properties

Output

A list of matching birds with descriptions

For example, your data might represent a bird like this:

```
bird = {

   "name": "Blue Jay",

   "color": "blue",

   "size": "medium",

   "habitat": "forest",
```

```
    "diet": "seeds"


}
```

You would store several birds in a list and loop through them to find matches. This is a great opportunity to demonstrate lists, dictionaries, conditionals, loops, functions, and data abstraction.

One important design decision

I'd suggest allowing users to leave some characteristics blank. For example, someone who knows a bird is blue but doesn't know its diet should still be able to search.

You could also make the search more flexible:

- Strict matching: Return birds matching every selected characteristic.
- Best matches: Rank birds by how many selected characteristics they match.

Start with strict matching because it's simpler to implement and debug. If you finish early,

you can add a best-match ranking system.

My recommendation

Keep the first version to around 20–40 birds with four searchable characteristics. That's enough to make the tool useful without requiring you to collect and manage an enormous amount of information.

One question that would help narrow down the project: Who do you picture using this bird search engine?

People who see a bird and want to identify it

People who want to discover and learn about birds

Birdwatchers looking for birds in a particular location

What birds should it cover?

Common birds in North America

Birds in a specific local area

Birds from around the world

Help me scope my Drafter CS1 bird search engine project. Intended audience: people who see a bird and want to identify it. Bird collection: common birds in North America. Recommend a realistic first-version feature set and project plan.

Project plan: North American Bird Finder

I recommend building a bird identification assistant that helps people narrow down which bird they saw by selecting its visible characteristics. The user doesn't need to know the bird's name or be an expert; they just need to remember a few details about its appearance.

For a three-week CS1 project using Drafter, this is a strong scope: it solves a clear problem, has a manageable amount of data, and demonstrates core programming concepts without requiring a backend or advanced AI.

1. Define the first version



The user's problem

"I saw a bird in my yard or on a walk, but I don't know what it was."



Your solution

The user chooses characteristics such as color, size, and habitat. The app searches a collection of common North American birds and displays possible matches.



The intended result

A short list of likely birds, with pictures and descriptions that help the user decide which one they saw.

The key is to describe the results as *possible matches*, not guaranteed identifications. Many birds look similar, and lighting, age, season, and plumage can make identification difficult.

2. Recommended feature set

Separate the essential features from optional improvements so that the project remains achievable.

Must-have features

Build these first. They make a complete application.

A simple home page explaining how to use the tool

Search form with dropdowns for bird color, size, and habitat

A collection of approximately 25–40 common North American birds

Python logic that filters birds based on selected characteristics

Results page showing matching bird names and short descriptions

A helpful message when no birds match, plus a way to reset the search

Nice-to-have features

Add these only after the basic search works.

A photograph for each bird, using reliable image URLs

A bird details page with fun facts and distinguishing features

A best-match ranking when no exact matches exist

Save favorite birds using browser local storage

Leave out of version one

- Identifying birds directly from uploaded photos

- AI-powered identification services
- Live bird sightings and migration tracking
- User accounts, a backend, or a cloud database
- Searching every bird species in the world

3. Choose the right search characteristics

Start with four characteristics at most. Each one should be something a person can reasonably observe without catching or handling the bird.

Characteristic	Example options	Why it helps
Main color	Red, blue, yellow, brown, black, white, gray	Often easy to notice
Size	Small, medium, large	Helps eliminate many species
Habitat	Backyard, forest, wetlands, grassland, coast	Narrows down possibilities

Beak or body clue	Long beak, short beak, crest, long tail	Helps distinguish similar birds

I would make the first three required features of the search system and treat the fourth as an optional enhancement. Allow users to skip unknown characteristics.

One design detail matters: a bird may have multiple colors and live in multiple habitats. Your data should support more than one color or habitat for each bird rather than storing only one value.

4. Build a manageable bird collection

Begin with approximately 25–40 species, choosing recognizable birds with a range of colors, sizes, and habitats.



American Robin

Medium · Orange, gray · Yards, parks



Blue Jay

Medium-large · Blue, white, black · Woodlands, yards

Northern Cardinal

Medium · Red or reddish-brown · Woodlands, yards



Black-capped Chickadee

Small · Black, white, gray · Forests, yards



Mourning Dove

Medium · Tan, gray, brown · Open areas, yards



Mallard

Medium-large · Green, brown, gray · Ponds, wetlands

These are examples, not a complete verified dataset. For your actual bird information, use a reputable reference such as the Cornell Lab of Ornithology

All About Birds

website. Verify characteristics and use images only when you have an appropriate source and permission to use them.

5. How the search should work

The core algorithm can be straightforward Python.

1. Store the birds in a list of dictionaries.
2. Read the user's selected characteristics.
3. Loop through the birds.
4. Compare each bird's properties against the selected options.
5. Add matching birds to a results list.
6. Display the results or a message if nothing matches.

For example, suppose a user selects blue, medium, and forest. The program checks every bird against those selections. A bird is included only if it satisfies all three selected criteria, using a consistent rule for matching multiple colors and habitats.

A useful improvement is to distinguish between *no match* and *no information*. If the user leaves habitat blank, the program shouldn't reject birds simply because the habitat wasn't selected.

6. A realistic three-week development plan

Week 1

Foundation

Build the data and search logic

- Define the bird data structure.
- Research and enter the first 15–20 birds.

- Write a function that compares characteristics.
- Test the function with different inputs, including blank fields and no matches.

Week 2

Interface

Connect the logic to Drafter

- Build the home page and search form.
- Connect dropdown selections to the filtering function.
- Display matching birds with their characteristics.
- Add reset-search and no-results behavior.
- Expand the collection toward 25–40 birds.

Week 3

Polish

Test, improve, and prepare the presentation

- Test unusual combinations and partial searches.
- Improve the layout and instructions.
- Add bird images if time allows.
- Fix bugs and verify the bird data.
- Prepare to explain your data structures, loops, conditionals, and search algorithm.

7. Suggested project goals for your worksheet

These draft answers align with the sections in your uploaded project planning worksheet.

CS1 Final Project Alpha F26.pdf

- Problem: People who see an unfamiliar bird may struggle to identify it using its appearance and surroundings.
- Solution: A web application that lets users select observable bird characteristics and returns a list of possible North American bird species.
- Core goal: Help users narrow down the identity of a bird by comparing their observations with stored bird information.

- Inputs: Color, size, habitat, and optionally distinctive physical features.
- Processing: Compare selected characteristics with the properties of each bird and filter the collection to find matches.
- Output: A list of possible bird species, with names, descriptions, and optionally pictures.
- Core data abstraction: A bird represented by its name, colors, size, habitats, and distinguishing features. Details irrelevant to identification can be omitted.
- Main technical challenge: Designing useful matching rules and collecting accurate, consistent data.
- AI strategy: Ask AI to explain small pieces of code, test generated code with sample birds, and make sure you can explain each function and conditional.

My final recommendation: Make the first version a *characteristic-based bird search tool*, not an automatic image recognizer. The search algorithm is achievable for a CS1 student, the result is genuinely useful, and optional pictures and best-match ranking give you room to improve it without risking the core project.

Part 8: Submit

In Google docs, with this document open, please click on *File* > *Download* > *Microsoft Word (.docx)*, and make sure you put that download somewhere you can find it again. Then, upload it to the Canvas assignment about selecting or brainstorming your final project.

When you're done with this assignment, review this page of the Drafter documentation to better prepare for the next step.